Investigating the Need for Closure Metrics in the Mid-Level Vision Toolbox

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Abstract

This paper investigates the need for and proposes new metrics for closure detection in the Mid-Level Vision (MLV) Toolbox, a computational framework for analyzing visual properties such as contours, orientation, and junctions (Walther et al., 2023). The effectiveness of existing metrics within the MLV Toolbox as metrics for the Gestalt principle of closure is investigated. The study finds that while current properties like contour length and the separation property of the medial axis transform (MAT) can give partial insights, they are insufficient for accurately modelling closure. Exploring more robust metrics to improve closure detection and quantification—such as the integration of contour completion and gap statistics—is recommended. Overall, this work aims to investigate and advance the current functionality of the MLV Toolbox, thereby aiding research in mid-level vision processes and current computational vision models.

Introduction

Gestalt principles of visual perception provide valuable insights into how the human brain organizes and interprets visual stimuli (e.g., objects and scenes). As proposed by Gestalt psychologists, the law of closure suggests that individuals perceive incomplete shapes or figures as complete based on the surrounding visual cues (Pinna & Deiana, 2015). This principle is fundamental in human object recognition, where the brain fills in gaps in fragmented visual information to form coherent objects (Gerhardstein et al., 2012).

The Mid-Level Vision Toolbox (MLV Toolbox) is a computational framework designed to analyze mid-level vision properties in images. It allows researchers to extract key visual properties such as orientation, length, curvature, and junctions and quantify MAT properties like symmetry and separation. The MLV Toolbox includes various functions to trace contour characteristics in computer-generated and artist-generated line drawings. Using the Segment Anything Model (SAM) (Kirillov et al., 2023), the toolbox can also generate line drawings from coloured images (Walther et al., 2023). These features make it a powerful tool for examining the mid-level perceptual processes involved in human vision.

The connection between Gestalt principles and visual processing is crucial for understanding human perception and improving machine perception in computer vision models. For example, a study by Pinna and Deiana (2015) showed that material properties (e.g., colour, shape, and depth) depend on the contours of objects in the scene. Another study by Gilden et al. (2012) found that natural contours, such as fractals and curves, have statistical properties that humans are instinctively attuned to, which aids in categorizing and recognizing scenes. This previous research suggests that including closure metrics within the MLV Toolbox could provide more avenues for research psychology.

The end goal of this study is to improve upon the current MLV Toolbox by addressing its need—or lack thereof—for closure metrics, furthering our understanding of mid-level vision processes, and improving existing computational vision models.

Closure in Literature

The use of neural networks for computational modelling to explain aspects of visual perception has had a long history in both computer science and psychology. Early work by Rumelhart, Hinton, and Williams (1986) introduced the back-propagation algorithm, which enabled neural networks to learn internal representations by minimizing error through iterative weight adjustments. This breakthrough helped networks develop hidden features that reflect patterns in visual data, forming the basis for later models that imitate how we see.

Building on these principles, Mozer (1991) extended the connectionist framework to model complex perceptual tasks, including recognizing multiple objects and allocating spatial attention. His model demonstrated that neural networks could simulate the grouping of visual elements, providing insight into how the brain might resolve ambiguities in incomplete images.

In parallel, early studies by Elder and Zucker (1993) laid the groundwork for the computational modelling of Gestalt principles by demonstrating that contour closure significantly influences the speed and accuracy of shape discrimination. Their work introduced a quantitative measure based on the sum of squares of the lengths of contour gaps—often called the L2 measure—which emphasizes significant gaps and captures a principle of perceptual regularity (Elder & Zucker, 1993; Elder & Zucker, 1994).

Subsequent research has explored closure in the context of neural networks. For instance, Kim et al. (2021) showed that neural networks trained on natural scenes show Gestalt closure effects, suggesting that exposure to structured visual data helps these systems develop closure-like processing. Other recent work investigating CNNs has revealed that conventional architectures struggle with fragmented visual inputs, prompting researchers to integrate closure-specific metrics to improve contour completion performance (Zhang et al., 2024).

Despite these advancements, closure remains an underdeveloped component in many computational vision frameworks. In computational models, existing methods for measuring closure often rely on indirect features like contour length (Elder & Zucker, 1996), curvature, and junction types (Ren et al., 2008). While these features help explain visual organization, they often miss the holistic nature of how we perceive closure. A more direct and robust set of closure metrics could significantly enhance the ability of computational models, such as the Mid-Level Vision (MLV) Toolbox, to detect and analyze image closure. Refining the representation of closure in these models could improve object recognition, edge detection, and scene interpretation tasks, bringing computational models of visual systems closer to human perceptual capabilities.

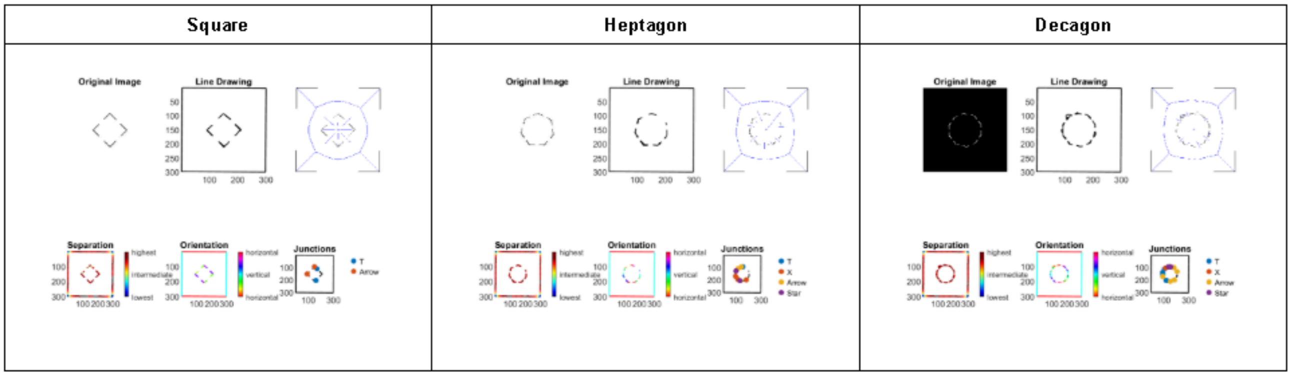

Figure 1: An example of the output from the MLV Toolbox process, consisting of six visualizations for each of three different shapes—a square, a heptagon, and a decagon. Each output includes the original image, the line drawing generated by the SAM, the medial axis transform (MAT), the separation heat map, the orientation heat map, and a plot of junctions.

Dataset

The dataset used to investigate the need—or lack thereof—for measures of closure within the MLV Toolbox was initially introduced by Kim et al. (2021) and further modified by Zhang et al. (2024) to investigate the Gestalt principle of closure in neural networks. This analysis only uses seven of the nine folders from the testing portion of the Zhang et al. dataset which comprises only aligned shapes with varying background colours, centre positions, and edge lengths. Each folder contains images of only one shape (triangle to decagon).

Each image measures 300×300 pixels and has different background colours (e.g. black or white), along with varying centre positions and edge lengths for the shape fragments. The edge lengths (defined as the length of the visible edge of each fragment) are set at 3, 8, 13, 18, 24, or 29 pixels. The dataset also standardizes the spacing between triangle vertices at 116 pixels to maintain consistent geometry (Kim et al., 2021; Zhang et al., 2024).

The dataset required no additional preprocessing, which maintained consistency with previous research and aligned it with known closure detection benchmarks. Its structured changes in position, size, and orientation made it a strong base for testing the MLV Toolbox's current closure detection abilities.

Methods

The dataset was directly integrated into the Mid-Level Vision (MLV) Toolbox before being processed by the SAM (Kirillov et al., 2023) into line drawings. These line drawings were then analyzed using the MLV Toolbox functions to extract relevant perceptual properties that could inform closure detection.

The computational process involved running each image through the SAM model to obtain contour-based representations, which the MLV Toolbox then processed. The toolbox's built-in functions calculated properties such as curvature and separation for closure analysis.

The output from the analysis consisted of a set of 6 visualizations for each image. These visualizations were the original image, the line drawing of the image (computed by the SAM), the MAT, the separation heat map, the orientation heat map, and a plot of junctions. Figure 1 displays an example of this output. Additionally, the toolbox generated an Excel sheet for each dataset folder containing quantified measurements of these properties for every image. These outputs provided qualitative and quantitative insights into the representation of closure within the dataset.

Properties Computed

In an attempt to compute metrics related to closure, several key properties were computed for each image using the MLV Toolbox. The following properties were extracted:

1. **Contour Length Statistics** (Mean, Median, Minimum, Maximum, Standard Deviation): Measures the lengths of individual contour segments, helping to assess whether closure can be quantified by contour length.

2. **Contour Length Histogram (per image):** A frequency distribution of contour lengths used to identify patterns in contour size and their role in forming closed shapes.

3. **Curvature Histogram (per image):** Captures how sharply contours bend, which can influence the likelihood of contours being grouped into a closed shape.

4. **Junction Histogram (per image):** Represents the count of different junction types (e.g., T-junctions, Y-junctions), which can indicate points where contours merge or diverge, which could indicate the presence or absence of closure.

5. **MAT Separation Means Statistics** (Mean, Median, Minimum, Maximum, Standard Deviation): Measures the distances between contour segments, helping to determine whether gaps between contours are small enough for closure to be inferred.

Metrics

When trying to derive meaningful metrics to quantify closure, a few approaches from the literature were explored. Unfortunately, many are outside the scope of the current functionality of the MLV toolbox.

One approach examined *contour length distributions*, where longer and more continuous contours were hypothesized to contribute to stronger closure perception (Elder & Zucker, 1996). Based on this hypothesis, contour length statistics were computed, and a histogram was generated; both were analyzed to evaluate their effectiveness as closure indicators.

Other investigations assessed whether *curvature or junction density* could provide insights into closure. Previous research by Ren et al. (2008) analyzed curvature statistics and junction distributions to determine their role in defining closed shapes within visual scenes. As a consequence of that research, curvature and junction histograms were examined to assess how well they indicated closure was present.

Using *gap statistics* to approximate endpoint proximity was also considered, as closure often emerges when contour endpoints are near one another (Elder & Zucker, 1994) and because the size and positioning of gaps influence whether observers perceive a shape as closed (Hu, 2024).

Unfortunately, computing gap statistics is not feasible with the current toolbox functionality, as there is no way to determine which contours belong to the same object in the image. Instead, inspired by a study on medial axis-based salience measures by Rezanejad et al. (2019), MAT separation means statistics were computed to study their ability to quantify closure.

As an extension, studies on *contour completion*, such as Rezanejad et al. (2021), have demonstrated how Gestalt grouping principles can aid in reconstructing incomplete shapes—their framework leveraged contour info,rmation to improve shape reconstruction in noisy environments. As with gap statistics, the toolbox does not currently have functionality related to contour completion.

Analysis & Results

Various metrics were examined to assess whether the computed properties could quantify closure, including contour length distributions, separation means, and initial considerations of junction and curvature properties. Contour length, curvature, and junction histograms were generated for each image, capturing the distribution of the properties for that image. However, interpreting these visualizations proved too complex to draw definitive conclusions.

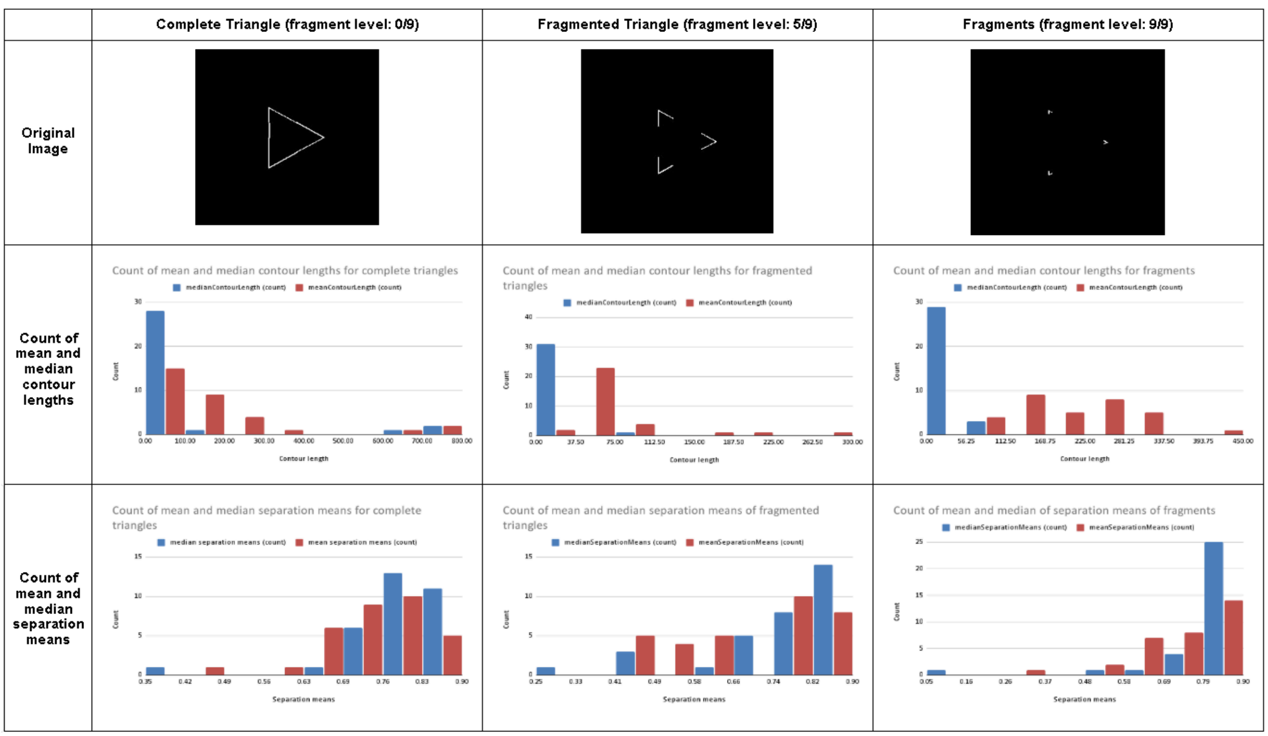

Figure 2: An example of the statistical analysis performed on the computed properties for a set of triangle images at different fragmentation levels. The figure examines the distribution of mean and median contour lengths across images at varying levels of fragmentation. Additionally, it analyzes separation means—a MAT property—showing that as fragment size decreases, the counts of mean and median separation values increase. This trend suggests that MAT separation means may serve as a closure metric.

Contour Length

Contour length statistics were analyzed to determine their potential as a closure metric. While maximum contour length was initially considered, it was confounded by the inclusion of image edge contours (which occurred during the conversion from the original image to the line drawing) and discarded. Instead, mean contour length, median contour length, and contour length histograms were examined across different levels of shape completeness (on a scale of 0-9, where 0 indicates completeness and 9 indicates the highest fragmentation level).

Figure 2 provides an example of the statistical analysis done on the computed properties for the set of triangle images. In particular, for contour length, the distribution of the mean and median across a set of images at a certain fragmentation level was examined. This analysis revealed no significant results—all median contour length distributions exhibited strong right skew (approaching zero as fragmentation level increased), while the means of contour lengths became more evenly distributed across shapes. This result suggests that dominant contours became shorter and more numerous as shapes became more fragmented. However, it is prudent to note that these findings may be affected by the compromised line drawing through the inclusion of image edge contours when converting the original image (as seen in Figure 1).

Junction and Curvature

While previous research has explored junction density and curvature statistics as indicators of closure, these properties were not extensively analyzed in this study. Initial visualizations (histograms) proved difficult to interpret, and further methodological refinement would be required to extract meaningful conclusions.

Separation Means

Separation means—a MAT property—showed slightly more promising results. The mean and median separation value counts increased as fragment size decreased, indicating that separation means increased as shapes became more fragmented (Figure 2). This outcome suggests a potential correlation between fragmentation and closure perception, albeit as a weak and indirect measure. While separation means may serve as a vague indicator of closure, further testing across different object types and conditions would be necessary to establish its validity as a metric.

Overall, while some properties showed slight relevance to closure detection, none provided a definitive or highly effective measure. These findings highlight the need for more robust metrics to improve the accuracy and reliability of closure assessment.

Limitations

Toolbox Limitations

The current implementation of the MLV Toolbox relies on the SAM (Kirillov et al., 2023) to generate line drawings of images before applying other toolbox functions. Although the model excels at processing human-made scenes and performs well, albeit slightly worse, with natural scenes, it encounters challenges in accurately detecting small fragments within images—especially when there is significant noise.

For example, an attempt to incorporate Doreen Hii and Zoran Pizlo's (2023) noisy image dataset-as described in their work on combining contour and region for closed boundary extraction-created challenges for the SAM encountered challenges in producing accurate and interpretable outlines from these noisy images (Figure 3). The inherent noise in the dataset disrupted the model's ability to detect and draw contours effectively, resulting in poor line drawings and, consequently, an inability to evaluate closure detection reliably.

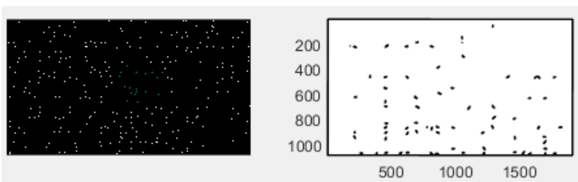

Figure 3: An example of an inaccurate line drawing produced from an image from the noisy image dataset from Hii and Pizlo (2023). Notice that some of the jitter has shaped into seed-like shapes while the rest of it is missing entirely.

Furthermore, when applying the toolbox to the primary dataset, issues were observed with some of the shape images, especially those with more complex or fragmented contours (Figure 4). For instance, shapes with smaller fragments (higher fragmentation level) failed to produce clean outlines due to the SAM's challenges with weak continuity cues. The results were incomplete or inaccurate shape reconstructions, compromising the evaluation of closure metrics in those particular instances.

Additionally, although the toolbox supports many perceptual properties, this study focused on a small subset of them highlighted in literature as potential metrics for closure. As a result, other relevant features, like symmetry or orientation, were not explored.

Dataset Limitations

The dataset used in this study is structured and well-curated but focuses on several shapes. This limits how well closure can be evaluated across a broader range of visual situations. Nevertheless, the dataset is generally suitable for studying closure, as controlled images are common in closure research. Exploring additional datasets would be helpful to better understand how closure works in complex scenes or object detection tasks.

Future Improvements

Future work could address toolbox and dataset limitations by expanding the range of visual properties investigated within the MLV toolbox and improving its ability to recognize closure patterns in fragmented or noisy conditions. Additionally, incorporating a wider variety of data, including visual scenes of different categories, would improve the robustness and generalizability of the toolbox's performance.

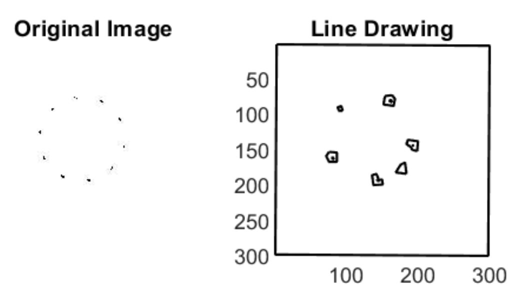

Figure 4: An example of an inaccurate line drawing of a decagon with a high fragmentation level. Notice that some fragments have become circles while others are missing entirely.

Recommendations

The statistical analysis and methods used in the MLV Toolbox were insufficient to fully quantify the principle of closure. While contour length, curvature, and junction density provided partial insights, they did not fully account for how humans perceive closed shapes.

Integrating *contour completion techniques* inspired by Rezanejad et al. (2021) into the MLV Toolbox is proposed to address these limitations. Extending incomplete contours based on contour and curvature properties can give a more accurate representation of closure. This extension would allow for a more holistic analysis of shape perception beyond what is captured by current toolbox properties.

Additionally, once contour completion is incorporated, *gap statistics* (Elder & Zucker, 1994) can be computed on the extended contours. These statistics would provide a quantitative measure of how discontinuities affect closure perception and refine the ability to distinguish between closed and open configurations.

To implement these metrics within the MLV Toolbox, existing functions should be adapted to support contour extrapolation and gap analysis. Visualization tools (e.g., complete contour overlay, histogram generation) should be updated to illustrate the effects of contour completion on closure perception. Future research should focus on validating these metrics against human perceptual judgments to ensure their effectiveness in modelling closure within computational vision frameworks.

Conclusion

This analysis examined the MLV Toolbox's ability to quantify the Gestalt principle of closure. The goal was to determine if the toolbox's existing metrics (such as contour length, curvature, junctions, and MAT separation) could effectively capture closure in images. While some measures, such as separation means, showed potential correlations with fragmentation levels, no single property provided a definitive or highly reliable quantification of closure.

The study faced many constraints regarding the limitations of the MLV Toolbox—its reliance on the Segment Anything Model (SAM) for contour extraction and its inability to compute gap statistics or perform contour completion. Challenges in interpreting curvature and junction histograms also pointed to the need for improvements in methodology. While the dataset was well-suited for controlled closure studies, it had a limited scope and lacked generalizability to real-world visual perception.

These findings emphasize the need to integrate direct closure metrics, such as contour completion and gap statistics, into the MLV Toolbox to improve its ability to quantify closure. Addressing these limitations would allow the toolbox to support research in mid-level vision processes better and advance current computational vision models.

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